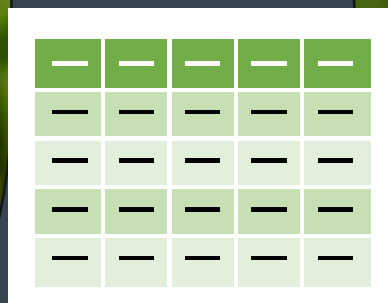






EXCEL

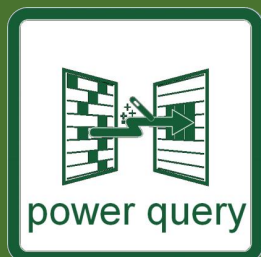
POWER QUERY
+
EXERCISE



Ludovic Moreau – moreau@artofnet.ch



EXCEL



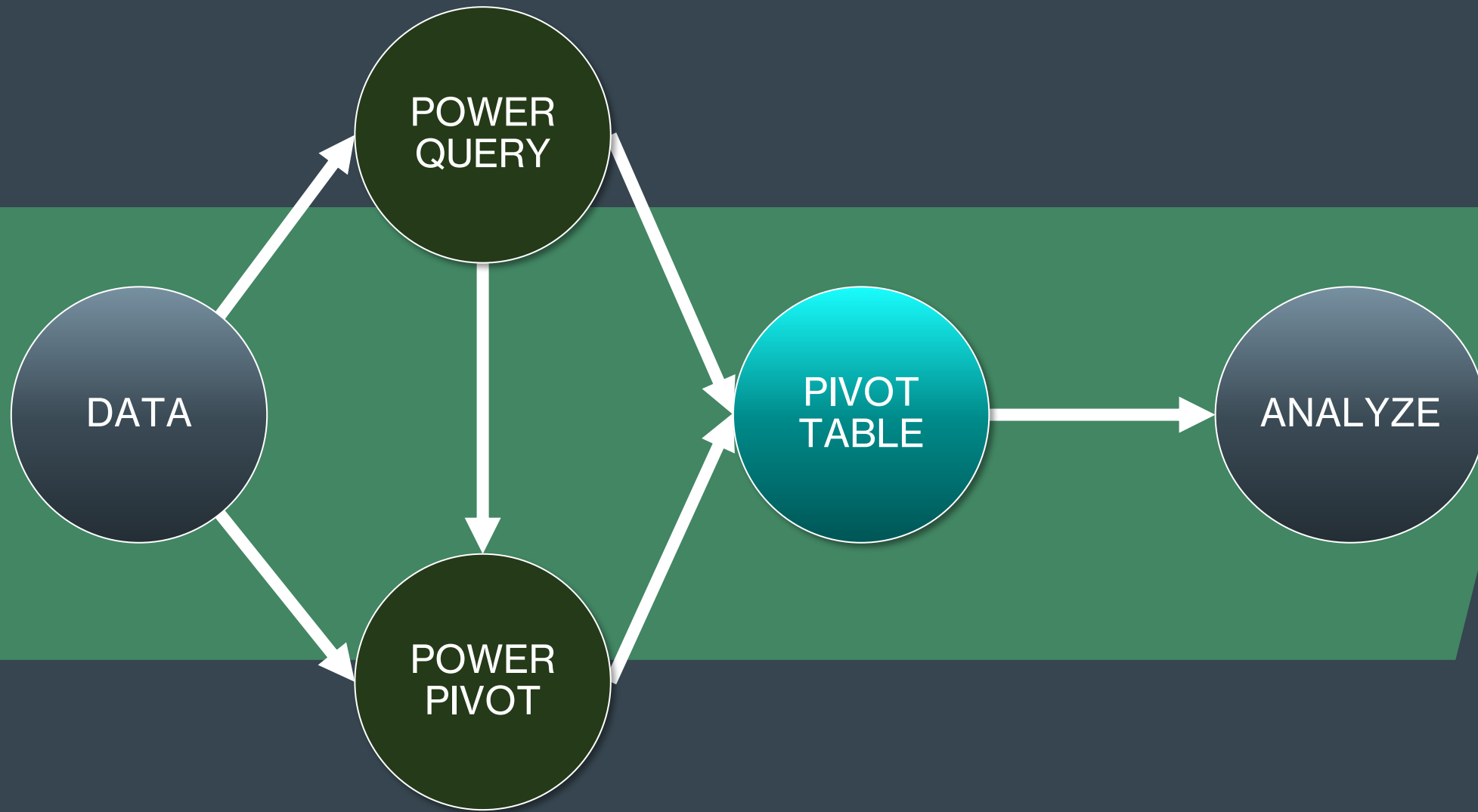
POWER
QUERY

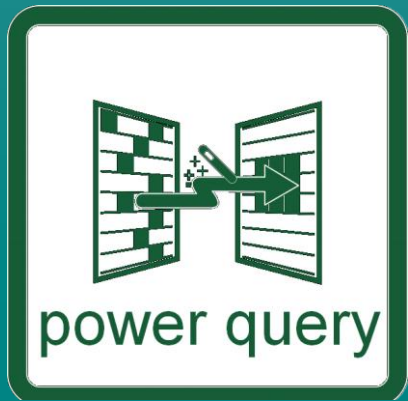


POWER
PIVOT



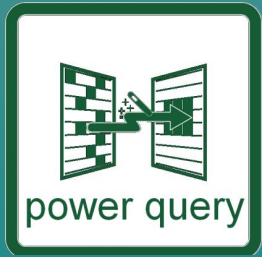
POWER
BI





POWER

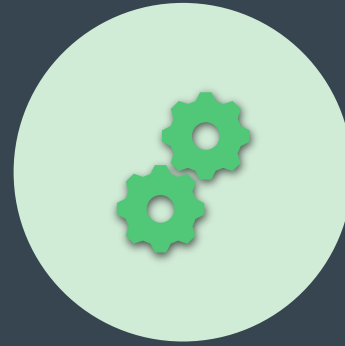
QUERY



ETL SOFTWARE



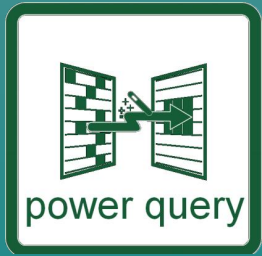
EXTRACTION



TRANSFORMATION



LOADING

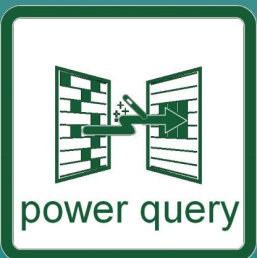


Key POINTS

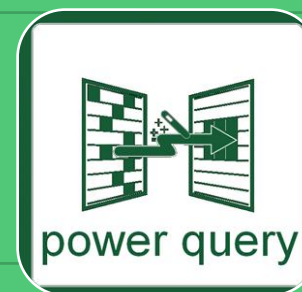
Cleaning and
visual
transformation

Reuse queries

Combine data
from multiple
sources



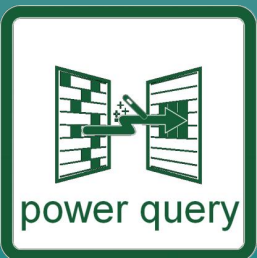
Import FROM MANY sources



Cleaning & Formatting

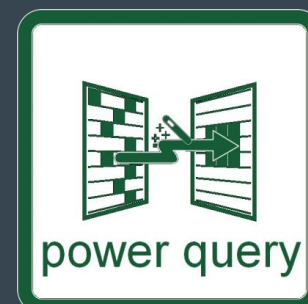
- Remove unnecessary rows and empty columns.
- Put the headers back at the top of the columns.
- Split a column by delimiter.
- Harmonize the type of data in each column.
- Import the data into EXCEL.

	Column1	Column2	Column3
1	Voici un tableau avec 50 noms de produits inform...	null	null
2	null	null	null
3	Noms de produit Poids (kg) Dimensions ...	null	null
4	----- ----- -----	null	null
5	Ordinateur portable 2.5 35 x 24 x 2...	null	null
6	Moniteur 5.2 60 x 40 x 1...	null	null
7	Clavier 0.8 45 x 15 x 3...	null	null
8	Souris 0.1 10 x 6 x 3 ...	null	null
9	Imprimante 7.8 45 x 35 x 2...	null	null
10	Scanner 3.2 30 x 20 x 1...	null	null
11	Routeur 0.4 20 x 15 x 5...	null	null
12	Modem 0.3 15 x 10 x 4...	null	null
13	Disque dur externe 0.2 10 x 8 x 2 ...	null	null
14	Cl? USB 0.02 5 x 2 x 0.5...	null	null
15	Carte graphique 1.2 25 x 10 x 4...	null	null
16	Carte m?re 0.6 30 x 20 x 3...	null	null
17	Processeur (CPU) 0.1 5 x 5 x 2 ...	null	null
18	M?moire RAM 0.05 8 x 2 x 0.5...	null	null
19	Disque dur interne 0.4 10 x 7 x 2 ...	null	null
20	SSD (Solid-State Drive) 0.05 5 x 3 x 0.5 ...	null	null
21	cran tactile 3.8 55 x 30 x 5 ...	null	null
22	Casque audio 0.3 20 x 18 x 1...	null	null
23	Haut-parleurs 1.2 25 x 15 x 1...	null	null
24	Webcam 0.1 5 x 5 x 2 ...	null	null
25	Microphone 0.2 10 x 5 x 3 ...	null	null
26	Tablette 0.5 25 x 18 x 1...	null	null
27	Smartphone 0.2 15 x 7 x 0....	null	null
28	Smartwatch 0.1 4 x 4 x 1 ...	null	null
29	couteurs sans fil 0.05 3 x 2 x 1 ...	null	null
30	Routeur Wi-Fi 0.3 15 x 10 x 4...	null	null
31	Adaptateur Bluetooth 0.02 3 x 2 x 1 ...	null	null
32	null	null	null
33		null	null
34	Batterie externe 0.2 10 x 5 x 2...	null	null



COMBINING

From a FOLDER

[illegible]

MERGE 2 QUERIES

SKU	souscatID
Valide 100%	Valide 100%
Erreur 0%	Erreur 0%
Vide 0%	Vide 0%
SKU-619955	5
SKU-367525	7
SKU-874528	16
SKU-167410	18
SKU-668353	25
SKU-772806	17
SKU-640119	29
SKU-154609	25
SKU-791801	35
SKU-129497	23
SKU-607728	41
SKU-239162	36
SKU-665945	16
SKU-894238	16
SKU-646568	27
SKU-249154	34
SKU-952912	16
SKU-438927	19
SKU-676138	26
SKU-642988	19
SKU-784630	42

Cost	Weight (kg)	SKU	Dimensions
Valide 100%	Valide 100%	Valide 100%	Valide
Erreur 0%	Erreur 0%	Erreur 0%	Erreur
Vide 0%	Vide 0%	Vide 0%	Vide
71	6.23	SKU-619955	17x18x14 cm
132	3.46	SKU-367525	43x36x4 cm
1309	7.28	SKU-874528	45x31x8 cm
954	8.67	SKU-167410	43x12x16 cm
703	2.77	SKU-668353	22x31x2 cm
834	6.65	SKU-772806	15x39x17 cm
230	5.51	SKU-640119	22x11x7 cm
1159	2.94	SKU-154609	35x22x7 cm
135	3.59	SKU-791801	59x10x10 cm
926	7.17	SKU-129497	51x33x8 cm
1241	7.9	SKU-607728	25x22x19 cm
1294	0.41	SKU-239162	54x13x18 cm
1440	2.12	SKU-665945	14x34x16 cm
1424	3.55	SKU-894238	56x25x8 cm
951	9.88	SKU-646568	18x21x17 cm
694	8.76	SKU-249154	48x10x17 cm
1073	6.63	SKU-952912	15x36x14 cm
453	7.76	SKU-438927	49x35x12 cm
839	0.78	SKU-676138	22x32x19 cm
1039	9.34	SKU-642988	43x37x14 cm
235	0.36	SKU-784630	28x14x2 cm
1054	4.7	SKU-968755	60x17x4 cm
1082	8.23	SKU-170209	19x33x7 cm
611	3.01	SKU-409176	52x16x15 cm
201	7.91	SKU-783285	50x21x13 cm
1346	8.94	SKU-353870	19x29x6 cm

MERGE 2 QUERIES

1 ² ₃ Cost	1.2 Weight (kg)	A ^B _C SKU	A ^B _C Dimensions	A ^B _C SKU	1 ² ₃ souscatID
● Valide 100% ● Erreur 0% ● Vide 0%	● Valide 100% ● Erreur 0% ● Vide 0%	● Valide 100% ● Erreur 0% ● Vide 0%	● Valide 100% ● Erreur 0% ● Vide 0%	● Valide 100% ● Erreur 0% ● Vide 0%	● Valide 100% ● Erreur 0% ● Vide 0%
71	6.23	SKU-619955	17x18x14 cm	SKU-619955	5
132	3.46	SKU-367525	43x36x4 cm	SKU-367525	7
1309	7.28	SKU-874528	45x31x8 cm	SKU-874528	16
954	8.67	SKU-167410	43x12x16 cm	SKU-167410	18
703	2.77	SKU-668353	22x31x2 cm	SKU-668353	25
834	6.65	SKU-772806	15x39x17 cm	SKU-772806	17
230	5.51	SKU-640119	22x11x7 cm	SKU-640119	29
1159	2.94	SKU-154609	35x22x7 cm	SKU-154609	25
135	3.59	SKU-791801	59x10x10 cm	SKU-791801	35
926	7.17	SKU-129497	51x33x8 cm	SKU-129497	23
1241	7.9	SKU-607728	25x22x19 cm	SKU-607728	41
1294	0.41	SKU-239162	54x13x18 cm	SKU-239162	36
1440	2.12	SKU-665945	14x34x16 cm	SKU-665945	16
1424	3.55	SKU-894238	56x25x8 cm	SKU-894238	16
951	9.88	SKU-646568	18x21x17 cm	SKU-646568	27
694	8.76	SKU-249154	48x10x17 cm	SKU-249154	34
1073	6.63	SKU-952912	15x36x14 cm	SKU-952912	16
453	7.76	SKU-438927	49x35x12 cm	SKU-438927	19
839	0.78	SKU-676138	22x32x19 cm	SKU-676138	26
1039	9.34	SKU-642988	43x37x14 cm	SKU-642988	19
235	0.36	SKU-784630	28x14x2 cm	SKU-784630	42

ADD QUERIES

position	commandeID	clientID	produitID	NB vente	date commande	date livraison
1	20-0001	19	DRO-786	117	04.01.2020	11.01.2020
2	20-0002	46	ACC-700	24	10.01.2020	16.01.2020
3	20-0003	27	AUD-836	2	11.01.2020	16.01.2020
4	20-0004	46	AUD-703	116	13.01.2020	19.01.2020
5	20-0005	16	SMR-766	129	15.01.2020	18.01.2020
6	20-0006	5	DRO-714	101	17.01.2020	22.01.2020

		position	commandeID	clientID	produitID	NB vente	date commande	date livraison
7	20-0007	1	21-0250	35	DRO-691	135	03.01.2021	06.01.2021
8	20-0008	2	21-0251	99	DRO-652	18	05.01.2021	12.01.2021
9	20-0009	3	21-0252	28	NET-343	21	08.01.2021	15.01.2021
10	20-0010	4	21-0253	90	SMR-668	139	09.01.2021	13.01.2021
11	20-0011	5	21-0254	83	NET-835	52	10.01.2021	13.01.2021
12	20-0012	6	21-0255	23	AUD-758	58	10.01.2021	15.01.2021
13	20-0013							

position	commandeID	clientID	produitID	NB vente	date commande	date livraison
1	22-0001	19	COM-384	31	08.01.2022	11.01.2022
2	22-0002	85	DRO-311	63	09.01.2022	14.01.2022
3	22-0003	31	SMR-317	57	10.01.2022	15.01.2022
4	22-0004	51	HAP-900	106	11.01.2022	14.01.2022
5	22-0005	32	NET-431	55	11.01.2022	16.01.2022
6	22-0006	61	SMR-887	41	11.01.2022	18.01.2022
7	22-0007	27	HAP-320	9	14.01.2022	17.01.2022
8	22-0008	31	SMR-342	113	15.01.2022	20.01.2022
9	22-0009	6	CAM-828	122	15.01.2022	19.01.2022
10	22-0010	92	WER-988	28	17.01.2022	20.01.2022
11	22-0011	81	DRO-835	6	18.01.2022	24.01.2022
12	22-0012	91	CAM-193	46	19.01.2022	26.01.2022

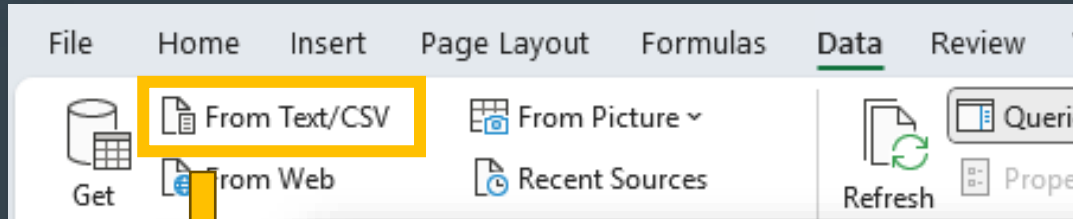
ADD QUERIES

commandeID	clientID	produitID	NB vente	date commande	date livraison
20-0001	19	DRO-786	117	04.01.2020	11.01.2020
20-0002	46	ACC-700	24	10.01.2020	16.01.2020
20-0003	27	AUD-836	2	11.01.2020	16.01.2020
20-0004	46	AUD-703	116	13.01.2020	19.01.2020
20-0005	16	SMR-766	129	15.01.2020	18.01.2020
20-0006	5	DRO-714	101	17.01.2020	22.01.2020
20-0007	36	SMR-672	41	19.01.2020	25.01.2020
20-0008	83	NET-279	31	19.01.2020	22.01.2020
20-0009	27	CAM-628	50	19.01.2020	24.01.2020
20-0010	18	COM-384	45	19.01.2020	25.01.2020
21-0254	83	NET-835	52	10.01.2021	13.01.2021
21-0255	23	AUD-758	58	10.01.2021	15.01.2021
21-0256	93	ACC-723	88	11.01.2021	17.01.2021
21-0257	7	ACC-997	96	12.01.2021	16.01.2021
21-0258	14	DRO-685	54	14.01.2021	20.01.2021
21-0259	79	AUD-871	146	17.01.2021	22.01.2021
21-0260	77	AUD-964	59	20.01.2021	26.01.2021
21-0261	90	AUD-964	52	20.01.2021	25.01.2021
22-0003	31	SMR-317	57	10.01.2022	15.01.2022
22-0004	51	HAP-900	106	11.01.2022	14.01.2022
22-0005	32	NET-431	55	11.01.2022	16.01.2022
22-0006	61	SMR-887	41	11.01.2022	18.01.2022
22-0007	27	HAP-320	9	14.01.2022	17.01.2022
22-0008	31	SMR-342	113	15.01.2022	20.01.2022
22-0009	6	CAM-828	122	15.01.2022	19.01.2022
22-0010	92	WER-988	28	17.01.2022	20.01.2022
22-0011	81	DRO-835	6	18.01.2022	24.01.2022
22-0012	91	CAM-193	46	19.01.2022	26.01.2022

POWER QUERY

EXERCISE

IMPORT from CSV



Computer_products_Sales(needs cleanup).csv

File Origin: 1252: Western European (Windows)
Delimiter: Comma
Data Type Detection: Based on first 200 rows

Column1
Name Weight Dimensions Price Units Sold Category
Cooling Fan Model 660 4.56 52x13x18 cm 1476.22 68...
Laptop Model 247 2.06 46x28x19 cm 1711.09 508 La...
Monitor Model 412 0.67 24x28x3 cm 184.81 460 Mo...
Processor Model 375 1.97 44x31x3 cm 134.08 852 P...
Switch Model 769 4.77 41x45x7 cm 372.68 29 Switch
Monitor Model 187 3.53 32x10x7 cm 501.11 534 Mo...
RAM Model 730 3.41 51x55x8 cm 1574.65 895 RAM
Switch Model 239 2.73 46x37x2 cm 1499.66 161 Swi...
Laptop Model 714 4.76 53x56x8 cm 892.38 897 Lapt...
Desktop Model 213 4.55 41x20x17 cm 1301.72 298 ...
HDD Model 570 1.01 35x51x20 cm 1176.91 209 HDD
Mouse Model 277 2.49 17x25x7 cm 1142.95 409 Mo...
Mouse Model 844 4.63 24x21x12 cm 415.8 105 Mou...
SSD Model 787 4.04 52x57x5 cm 710.26 565 SSD
Printer Model 165 3.19 57x30x17 cm 1518.97 322 Pr...
SSD Model 226 2.9 41x53x8 cm 332.16 619 SSD
RAM Model 219 1.01 41x39x17 cm 631.04 903 RAM
SSD Model 256 2.17 46x44x5 cm 20.47 318 SSD
Mouse Model 649 1.24 13x39x13 cm 1575.31 578 M...
Processor Model 491 4.71 43x13x3 cm 1401.2 837 P...
Cooling Fan Model 195 3.38 25x30x19 cm 1552.67 20...

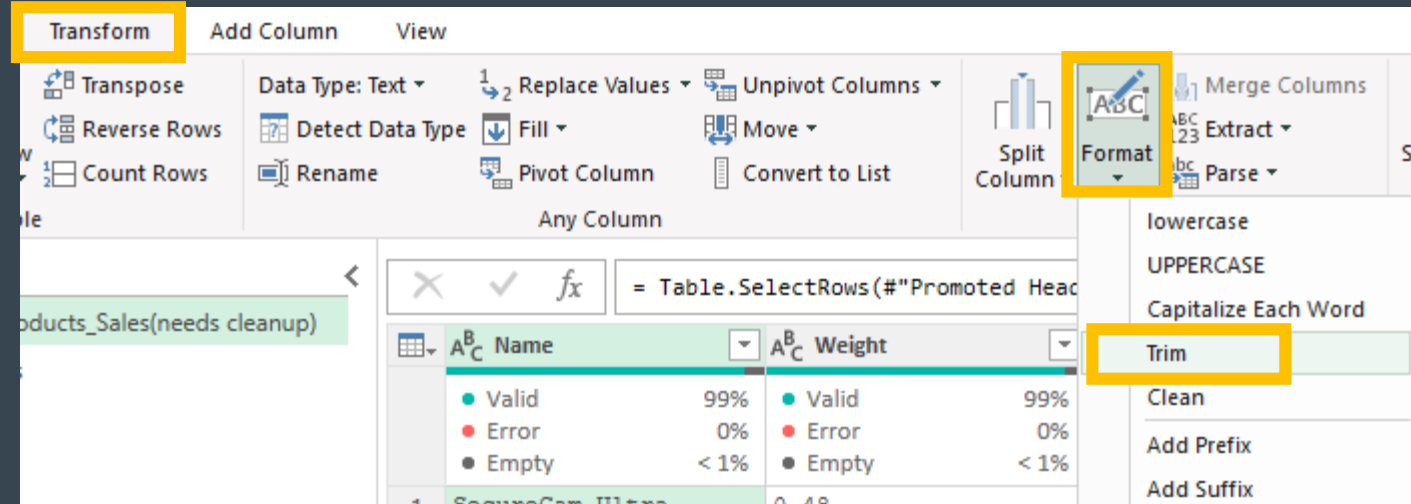
Load Transform Data Cancel



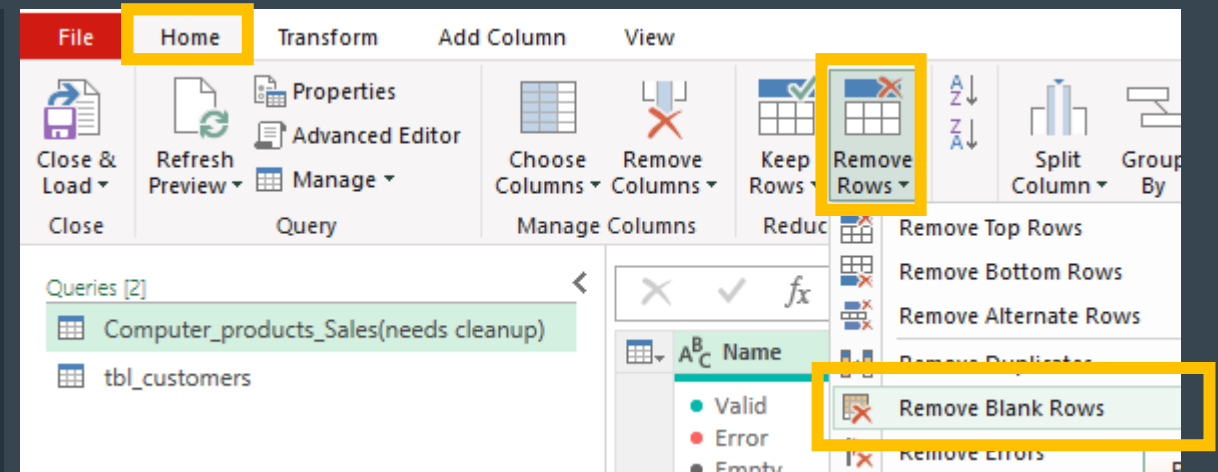
Column1
Valid 100%
Error 0%
Empty 0%
1 Name Weight Dimensions Price Units Sold Category
2 Cooling Fan Model 660 4.56 52x13x18 cm 1476.22 687 Cooling Fan
3 Laptop Model 247 2.06 46x28x19 cm 1711.09 508 Laptop
4 Monitor Model 412 0.67 24x28x3 cm 184.81 460 Monitor
5 Processor Model 375 1.97 44x31x3 cm 134.08 852 Processor
6 Switch Model 769 4.77 41x45x7 cm 372.68 29 Switch
7 Monitor Model 187 3.53 32x10x7 cm 501.11 534 Monitor
8 RAM Model 730 3.41 51x55x8 cm 1574.65 895 RAM
9 Switch Model 239 2.73 46x37x2 cm 1499.66 161 Switch
10 Laptop Model 714 4.76 53x56x8 cm 892.38 897 Laptop
11 Desktop Model 213 4.55 41x20x17 cm 1301.72 298 Desktop
12 HDD Model 570 1.01 35x51x20 cm 1176.91 209 HDD
13 Mouse Model 277 2.49 17x25x7 cm 1142.95 409 Mouse
14 Mouse Model 844 4.63 24x21x12 cm 415.8 105 Mouse
15 SSD Model 787 4.04 52x57x5 cm 710.26 565 SSD

CLEAN TEXT COLUMNS AND REMOVE BLANKS

The Name column contains undesirable spaces, clear these with Transform > Format > Trim

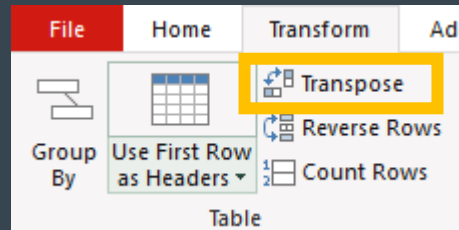
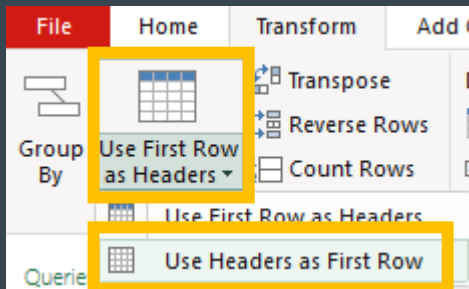


Remove blank rows by clicking Home > Remove Rows > Remove Blank Rows



Merging the content of a split row

- ▶ Row 19 was split into 2 separate rows (19+20). In order to merge the cells back together, transpose the table, merge columns 19 & 20 then transpose back.
- ▶ First step is to convert the headers into the first row so that we don't lose them in the transpose step. Once the transpose & Merge is over, promote the 1st as headers.



This screenshot shows the 'Transform' ribbon in Power Query. The 'Merge Columns' button is highlighted with a yellow box.

The formula bar shows the following formula:

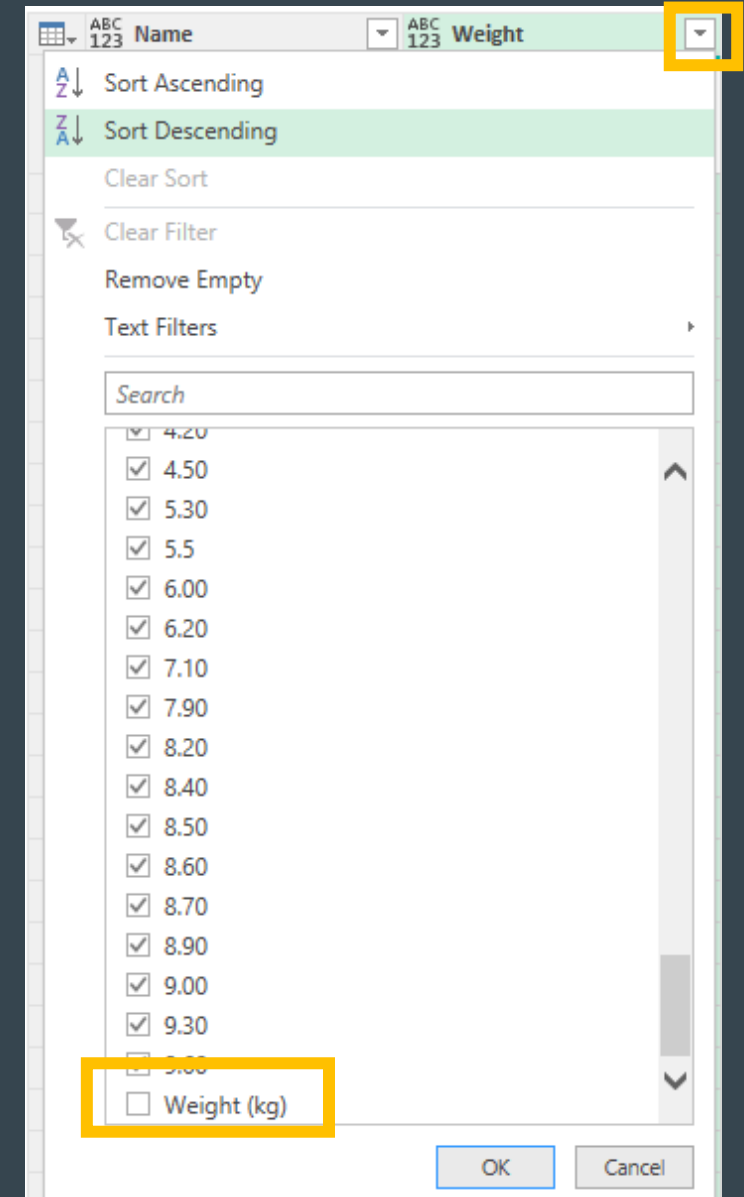
```
= Table.Transpose(#"Demoted Headers (not to lose them)")
```

The table below shows the data after the transpose operation. The columns are labeled 'Column19', 'Column20', and 'Column21'. The rows are numbered 1 through 8.

	Column19	Column20	Column21
1	SpeedSSD 1.5TB		SpeedSSD 500G
2	0.08		0.08
3	10x7x1 cm		10x7x1 cm
4		129.99	59.99
5		25	243
6		SSD	SSD
7		46127	2026-06-08
8		C299	C128

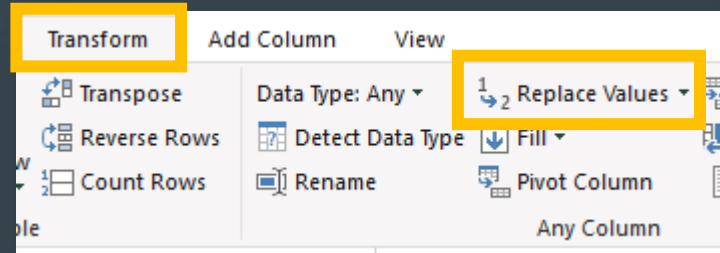
Getting rid of multiple headers

- ▶ Row 30 contains a copy of the headers. Filter it out by unchecking the 'Weight (kg)' value in the [Weight] column.
- ▶ Filtering will remove hidden rows in the final dataset.



Replacing incorrect date value

- ▶ Row 18 contains a date in a different format than the other rows. The problem comes from the original CSV file. You can either correct the value in the CSV file or replace the value in Power query by using Transform > Replace > Values.



Replace Values

Replace one value with another in the selected columns.

Value To Find

Replace With

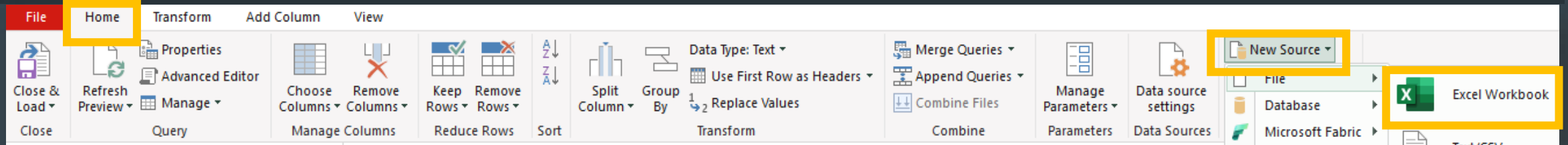
▶ Advanced options

OK Cancel

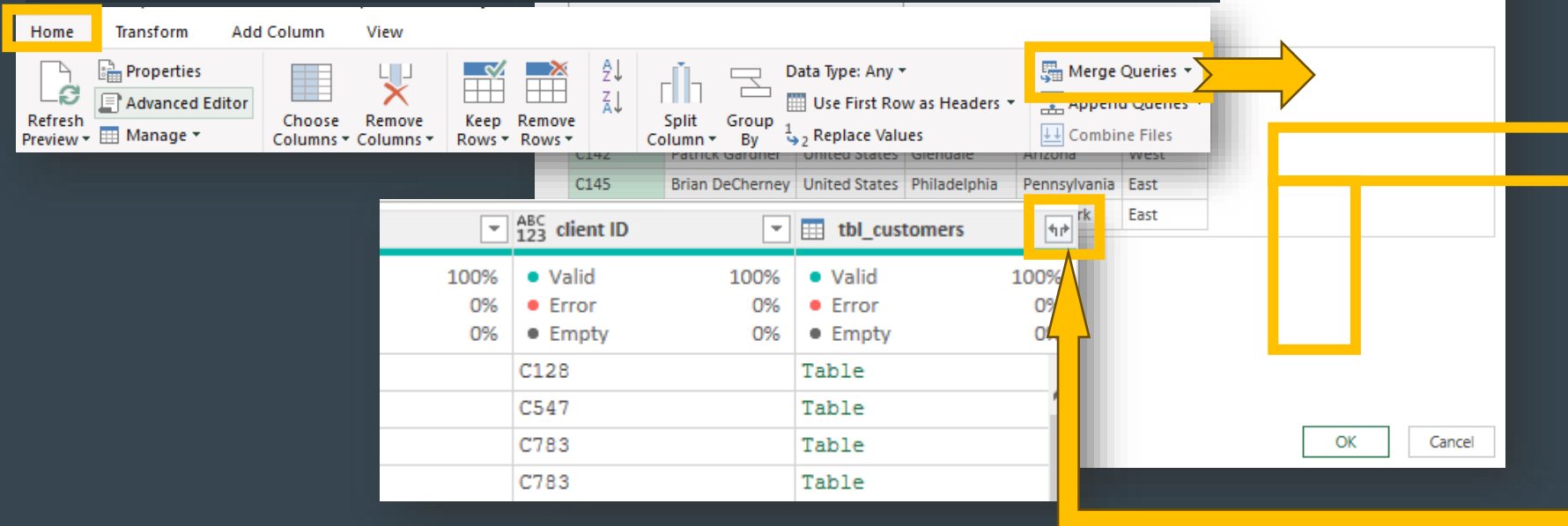
- ▶ Similarly, replace the decimal point in the [Weight] and [Price] columns by the appropriate symbol on your computer , the comma for example (if needed because your windows environment might already use the '.' as a decimal separator).

Merging with the customer info

- ▶ Create a new query on the Customers.xlsx file

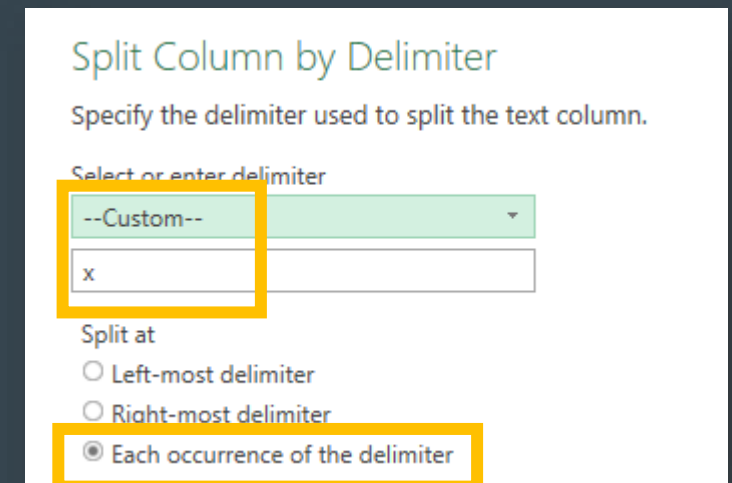
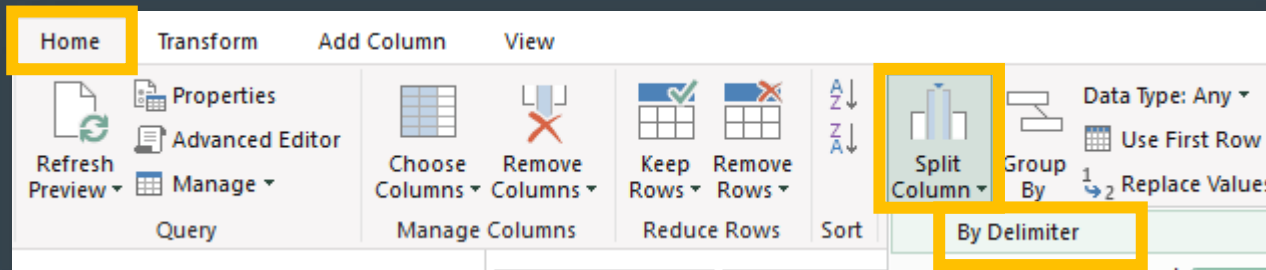


- ▶ Merge both queries based on the [client ID] column and expand the merged columns



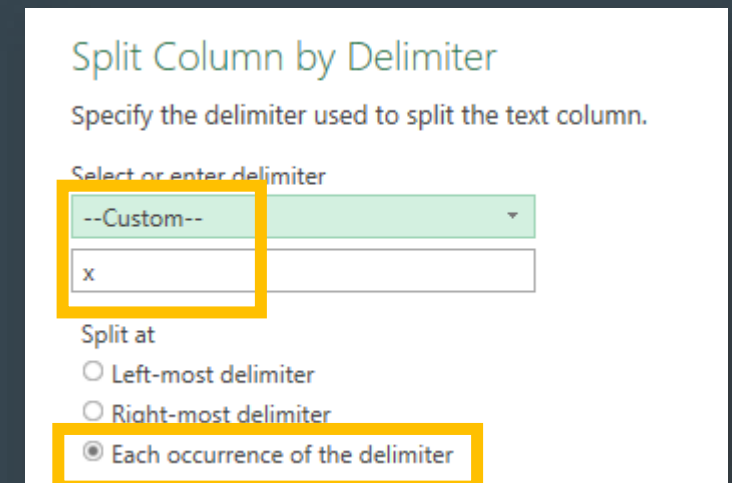
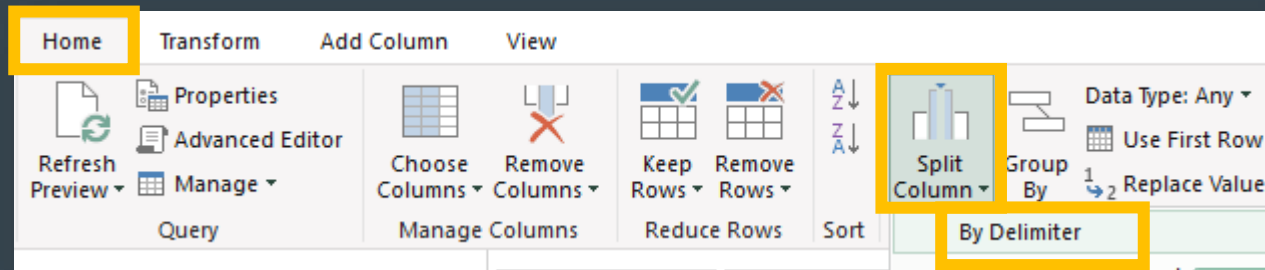
Create the dimensions columns

- The [Dimensions] column contains 3 pieces of information that could be useful for analysis or filtering.
- Convert this unique column in 3 new columns [Length], [Width] and [Height] by using Home > Split Column > Split by delimiter. Rename the columns newly created.



Create the dimensions columns

- The [Dimensions] column contains 3 pieces of information that could be useful for analysis or filtering.
- Convert this unique column in 3 new columns [Length], [Width] and [Height] by using Home > Split Column > Split by delimiter. Rename the columns newly created.

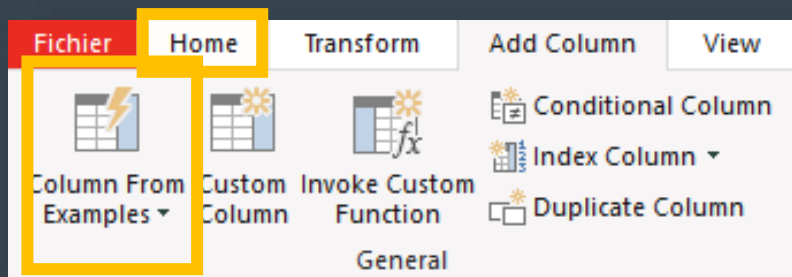


Create a column by example

- Get rid of the 'cm' values in [Dimensions.3] by adding a column from example.

Add Column > Column From Examples > From Selection (having selected [Dimensions.3])

- Delete the original [Dimensions.3] and move the newly created column next to the other dimensions cols and manually rename all three [Length], [Width] and [Height].
- Replace the '.' by ',' in [Height] (if needed) then convert to decimal type.

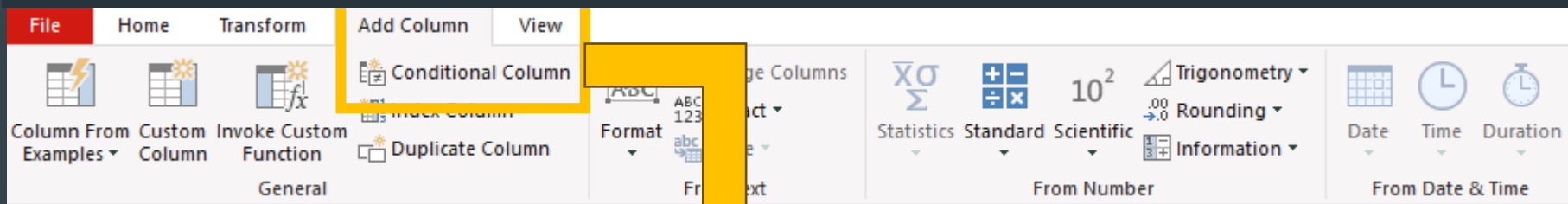


The image shows a data table with four columns: 'Dimensions.3', 'Price', 'Qty', and 'Text Before Delimiter.1'. The 'Text Before Delimiter.1' column is highlighted with a yellow box, showing the values extracted from the 'Dimensions.3' column. The table has a header row and several data rows. The 'Text Before Delimiter.1' column is currently empty, indicating it has been created but not yet populated with data.

Dimensions.3	Price	Qty	Text Before Delimiter.1
7 cm	169,99	82	7
1 cm	59,99	243	1
3 cm	34,99	4	3
43 cm	849,99	41	43
2.5 cm	22,99	196	2.5
2.5 cm	22,99	196	2.5
8 cm	49,99	182	8
42 cm	799,99	18	42

ADD A CONDITIONAL COLUMN

- Create the [Weight cat] column by adding a conditional column Add Column > Conditional Column then type the conditions as seen below:



Add Conditional Column
Add a conditional column that is computed from the other columns or values.

New column name

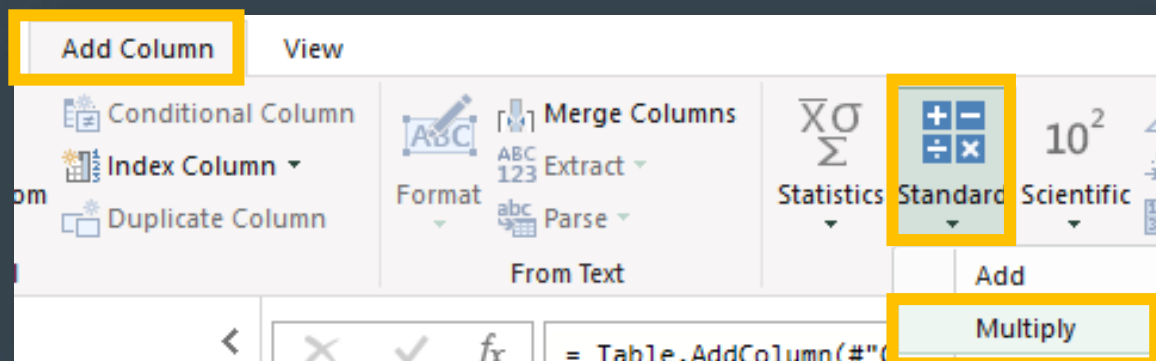
	Column Name	Operator	Value ①		Output ①
If	Weight		ABC 123 1	Then	ABC 123 Light
Else If	Weight		ABC 123 2	Then	ABC 123 Medium
Else If	Weight		ABC 123 3	Then	ABC 123 Heavy

Else ①
ABC 123 Very Heavy

1.2 Weight	Weight cat
0.48	Light
0.08	Light
0.7	Light
8.7	Very Heavy
0.29	Light
0.29	Light
0.27	Light

ADD THE [SALE AMOUNT] COLUMN

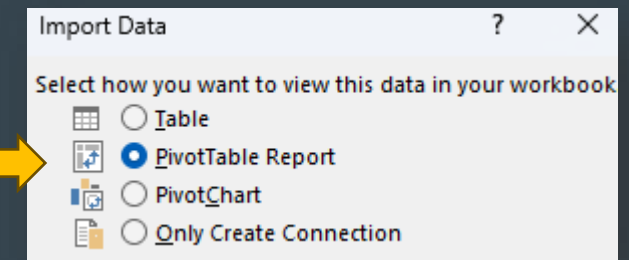
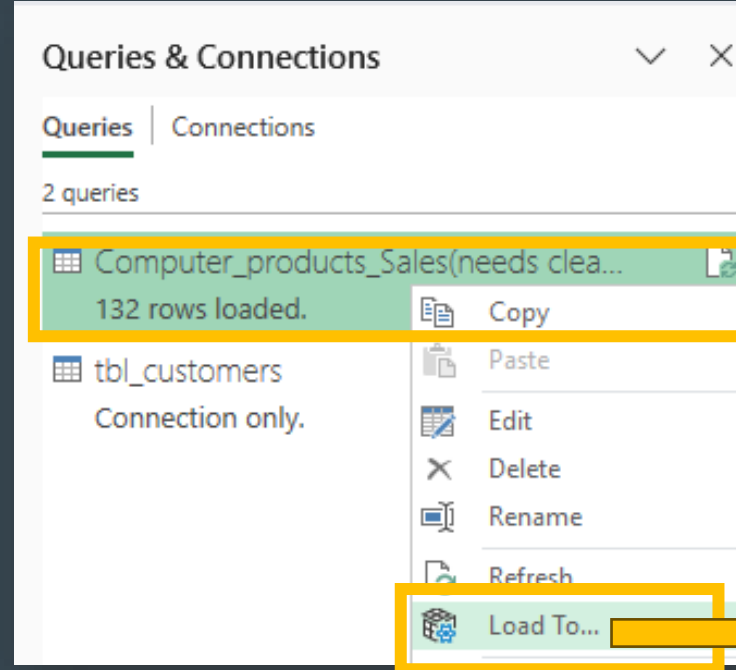
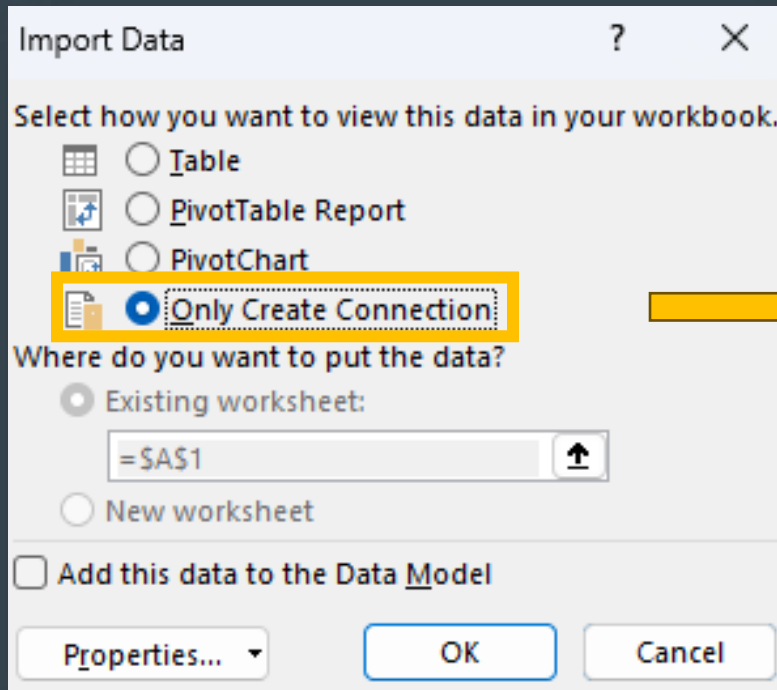
- We want to insert a column that calculates the amount of each sale.
- Select [Qty] & [Price] then click on Add Column > Standard > Multiply.
- Rename the newly created column [Sale amount]

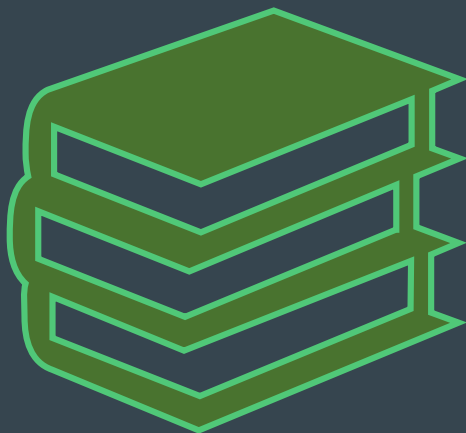


1.2 Sale amount
13939.18
14577.57
139.96
34849.59
4506.04
4506.04
9098.18
14399.82
36499.27
33669.09
12808.17
1132

IMPORT THE DATA IN EXCEL

- Once the cleanup is finished, click on Home > Close & Load > Close & Load To.
- Then right click on the Computer products query in the right pane and select 'Load To'. In the popup window, choose 'PivotTable Report'.





- Power Query in Excel
<https://support.microsoft.com/en-us/excel/about-power-query-in-excel>
- M language Reference (Power query):
<https://learn.microsoft.com/en-en/powerquery-m/power-query-m-function-reference>

Votre feedback

<https://learn.digicomp.ch>

<https://digicomp.my.salesforce-sites.com/EKBF?lang=fr>

